

Ujjwal Patel

+1 (469) 494-2266 | ujjwalpersonal09@gmail.com | [linkedin.com/in/ujjwal-patel09](https://www.linkedin.com/in/ujjwal-patel09) | github.com/ujjwalpatel09

WORK EXPERIENCE

Devai Technologies

AI/ML Engineer

Pune, India

Jul 2021 - Jul 2024

- Developed a real-time telematics feature store using Apache Spark and Snowflake, ingesting 150K+ behavioral records daily and trimming data processing overhead by 40% while improving downstream model accuracy.
- Fine-tuned EfficientNet-based computer vision models through 30+ systematic ablation studies, achieving 84.12% classification accuracy across 200 categories and surpassing the pre-training baseline by 12%.
- Engineered an automated ML retraining system with Airflow and MLflow, scheduling drift-triggered retraining, versioning experiment runs, and eliminating 35% of model degradation incidents across 24 release cycles.
- Constructed an end-to-end model evaluation framework with A/B testing, anomaly alerting, and precision, recall, and F1 tracking across 8 KPIs, eliminating 35% of post-deployment failures before reaching production.
- Deployed a transformer-based NLP classification platform on AWS with Kubernetes, serving text extraction and intent classification models at sub-300ms latency with 99%+ uptime.
- Built production LLM-powered pipelines using LangChain and RAG architectures from 2023 to 2024, driving task error rate from 18% to under 4% while processing 1,000+ requests daily in production.
- Shipped ML infrastructure tooling across Docker, Kubernetes, and CI/CD pipelines, establishing consistent deployment standards and improving sprint delivery velocity by 30%.

PROJECTS

MealMind

Personal Project

- Designed a 3-stage Claude Sonnet vision pipeline (Python, Streamlit) turning a fridge photo into 3 ranked recipes in under 10s, raising recipe feasibility from 41% to 87%.
- Built ingredient recognition reaching 89% inventory-use accuracy by enforcing Pydantic structured schemas and a 5-bucket quantity calibration step, cutting hallucinated ingredients ~60%.
- Orchestrated a multi-objective ranking function plus a skip/made-it preference loop, lifting the user "would make this" rate from 38% to 74%.

MeetingMemory

Personal Project

- Architected a hybrid graph + vector RAG system (NetworkX, FAISS, Claude Sonnet) over meeting transcripts, reaching 89% answer accuracy with a query router that delivered a 2x quality gain over pure vector RAG.
- Instrumented structured Pydantic extraction with ingestion-time source spans, hitting 91% decision recall and 94% citation precision at 2.4s p50 query latency.
- Documented design tradeoffs across extraction, routing, and pairwise contradiction detection (82% precision), improving accuracy and reproducibility.

Bird Species Classification - Deep Learning

Personal Project

- Implemented EfficientNet with transfer learning for 200-class fine-grained visual recognition across approx. 20K images, completing 30+ ablation runs with 5-fold cross-validation to achieve 84.12% accuracy, 12% above baseline.
- Scaled training infrastructure on AWS with distributed compute and Docker CI/CD, applying class-imbalance corrections that lowered model variance by 18% and trimmed environment setup time by 60%.
- Established reproducible ML delivery standards by versioning data, model artifacts, and experiment metadata across 8 iterations with MLflow and audit-ready documentation.

EDUCATION

University of Texas at Arlington

Master of Science in Computer Science (In Progress)

TX, USA

Graduation Date: May 2026

GPA: 3.5/4.0 | Coursework: Machine Learning, Artificial Intelligence, NLP, Distributed Systems, Cloud Computing

Mumbai University

Bachelor of Science in Computer Science

India

Graduation Date: May 2023

GPA: 3.8/4.0 | Coursework: Statistics, Data Analysis, Computer Vision, Data Mining, NLP

SKILLS & INTERESTS

Programming Languages	Python, SQL, R, Bash/Shell, C++, Java
Machine Learning & AI	Supervised Learning, Unsupervised Learning, Deep Learning, Computer Vision, NLP, Generative AI, Large Language Models, Reinforcement Learning, Multi-agent Systems
Frameworks & Libraries	PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Hugging Face Transformers, OpenCV, XGBoost, FastAPI, spaCy, Keras
MLOps & Deployment	MLflow, Airflow, Docker, Kubernetes, AWS SageMaker, Azure ML, Databricks, CI/CD Pipelines, GitHub Actions, GitLab CI/CD
Data Engineering & Visualization	pandas, NumPy, Apache Spark, Snowflake, SQL, Power BI, Tableau, Matplotlib, Seaborn, dbt
Cloud & Tools	AWS EC2, AWS S3, AWS SageMaker, AWS Lambda, GCP Vertex AI, GCP BigQuery, Azure, Terraform, Linux/Unix, Bash Scripting
Certifications	Oracle Cloud Infrastructure OCI 2025 Certified Generative AI Professional and AI Foundations Associate Google Cloud Facilitator Program - 13+ Skill Badges
Soft Skills	Agile Methodologies, Technical Documentation, Code Reviews, Stakeholder Communication, Root Cause Analysis, Team Collaboration